Physicochemical Differences, Dimensionality, and Latent Structure of Portuguese Vinho Verde Wines: A MANOVA, Principal Component, and Exploratory Factor Analysis of the UCI Wine Quality Dataset

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2. Introduction

Wine quality is ultimately judged by sensory panels, but the liquid those panels taste is a chemical system whose composition can be measured objectively and cheaply. Routine laboratory measurements of acidity, sugar, sulfur dioxide, alcohol, and related constituents are used by producers and regulators to monitor stability, safety, and style. Understanding how these measurements differ between wine types, how many independent dimensions they actually carry, and what latent chemical processes generate their correlations is useful both scientifically and practically, because redundant measurements can be consolidated and type-specific profiles can be identified (Cortez et al., 2009).

This report analyses the physicochemical profiles of red and white Portuguese *Vinho Verde* wines with three complementary multivariate procedures. A multivariate analysis of variance (MANOVA) asks whether the two wine colours differ in their chemistry when several characteristics are considered simultaneously. Principal component analysis (PCA) asks how far the 11 measurements can be compressed into a few composite dimensions. Exploratory factor analysis (EFA) asks which latent chemical dimensions best explain the correlations among the measurements. Sensory quality is deliberately excluded from all three analyses, as required by the assessment.

3. Dataset

The Wine Quality dataset (Cortez et al., 2009) is hosted by the UCI Machine Learning Repository (dataset ID 186; https://archive.ics.uci.edu/dataset/186/wine+quality; licensed CC BY 4.0). It contains laboratory results for Portuguese *Vinho Verde* wines from the Minho region: **1,599 red** and **4,898 white** samples, **6,497** wines in total. Each wine has 11 continuous physicochemical measurements, an integer sensory quality score (median of at least three expert evaluations), and, after the two files are combined, a categorical *color* variable (Red or White). Table 1 lists the variables and their units.

Table 1

Variables in the Wine Quality dataset

Variable	Type	Unit	Role in this study
fixed acidity	Continuous	g/dm³ (as tartaric acid)	PCA/EFA (all 11); MANOVA (six selected)
volatile acidity	Continuous	g/dm³ (as acetic acid)	PCA/EFA (all 11); MANOVA (six selected)
citric acid	Continuous	g/dm³	PCA/EFA
residual sugar	Continuous	g/dm³	PCA/EFA (all 11); MANOVA (six selected)
chlorides	Continuous	g/dm³ (as sodium chloride)	PCA/EFA (all 11); MANOVA (six selected)
free sulfur dioxide	Continuous	mg/dm³	PCA/EFA
total sulfur dioxide	Continuous	mg/dm³	PCA/EFA (all 11); MANOVA (six selected)
density	Continuous	g/cm³	PCA/EFA
pH	Continuous	pH units	PCA/EFA
sulphates	Continuous	g/dm³ (as potassium sulphate)	PCA/EFA (all 11); MANOVA (six selected)
alcohol	Continuous	% vol.	PCA/EFA
quality	Integer (3–9 observed)	Sensory score	Not analysed
color	Categorical	Red / White	MANOVA grouping variable

Note. Units follow the UCI documentation.

Data provenance. The analysis script imports the data with `ucimlrepo.fetch_ucirepo(id=186)` and, if the repository cannot be reached, falls back to the two original UCI files (`winequality-red.csv` and `winequality-white.csv`) stored in the repository's `data/` folder. The results reported here were generated from those two CSV files, which contain the expected 1,599 red and 4,898 white records.

4. Research Questions and Hypotheses

RQ1 (MANOVA). Do red and white wines differ significantly in their overall physicochemical characteristics when several characteristics are considered simultaneously?

H_0: the mean vector of the six selected physicochemical variables is the same for red and white wines ($\boldsymbol{\mu}_{red} = \boldsymbol{\mu}_{white}$). H_1: the mean vectors differ. Decisions use $\alpha = .05$; follow-up univariate tests use a Bonferroni-adjusted $\alpha = .05/6 = .0083$.

RQ2 (PCA). Can the 11 physicochemical measurements be summarised by a smaller number of principal components that retain a substantial proportion of total variability?

PCA is a descriptive technique with no formal null hypothesis, but its usefulness depends on the variables being correlated. This is checked with Bartlett's test (H_0: the population correlation matrix is an identity matrix; H_1: it is not) and the Kaiser–Meyer–Olkin (KMO) measure.

RQ3 (EFA). What underlying latent dimensions explain the relationships among the physicochemical characteristics of wine?

The same Bartlett hypotheses apply (H_0: no common variance to model). The substantive expectation is that a small number of latent chemical dimensions (for example, preservation,

body/density, and acidity) generate the observed correlations.

5. Methodology

Software. All analyses were run in Python 3.12.3 with pandas 3.0.2, NumPy 2.4.4, SciPy 1.17.1, statsmodels 0.15.0, scikit-learn 1.8.0, pingouin 0.7.0, factor_analyzer 0.5.1, matplotlib 3.10.8, and seaborn 0.13.2. The random seed is 42, so every simulation-based result (parallel analysis, permutation test) is reproducible.

Data preparation and exploratory analysis. The red and white files were imported, a *color* label was attached, and the files were concatenated (n = 6,497). I checked structure, missing values, and exact duplicate rows; computed descriptive statistics (mean, SD, quartiles, skewness, excess kurtosis, coefficient of variation); screened univariate outliers with Tukey's 1.5 × IQR and $|z| > 3$ rules; and screened multivariate outliers with Mahalanobis D^2, computed within colour and compared with the χ^2 quantile at $p = .001$. Pearson correlations were examined with a heatmap and a scatterplot matrix.

MANOVA. The grouping variable was colour. Six dependent variables were chosen from the exploratory results (Section 6.2). Assumptions were assessed through study design (independence), Mardia's multivariate skewness and kurtosis tests and χ^2 Q–Q plots (normality), within-group Mahalanobis distances (outliers), Box's M with Brown–Forsythe tests and a comparison of covariance determinants (homogeneity), and within-group correlations and variance inflation factors (relationships among the dependent variables). All four multivariate statistics were computed with statsmodels; Pillai's trace was designated the primary statistic because it is the most robust to violations of normality and covariance homogeneity (Olson, 1976). Effect size is partial η^2 ($= 1 - \text{Wilks' } \Lambda$) and the Mahalanobis distance between group centroids. Robustness was examined with a 10,000-permutation test and re-analyses after log transformation, outlier removal, duplicate removal, and balancing the groups. Follow-ups comprised univariate ANOVAs with η^2, Cohen's d, Welch's F, and a discriminant function (standardised coefficients, structure correlations, 10-fold cross-validated accuracy).

PCA. PCA was applied to the correlation matrix of the 11 z-standardised variables (colour and quality excluded). Retention used eigenvalues, the scree plot, proportion and cumulative variance, Horn's parallel analysis (1,000 random normal data sets, 95th percentile), and the broken-stick criterion. Loadings are eigenvector coefficients scaled by the square root of the eigenvalue, i.e., correlations between variables and components.

EFA. Factorability was assessed with the correlation matrix, KMO, and Bartlett's test. The number of factors was judged from the common-factor scree plot, parallel analysis on the reduced correlation matrix, Velicer's minimum average partial (MAP) test, residual fit (RMSR, maximum-likelihood RMSEA/TLI), and interpretability. Factors were extracted by minimum residual (minres; an ordinary-least-squares common-factor method that does not require multivariate normality) and rotated with direct oblimin.

6. Results

6.1 Exploratory Data Analysis

Structure and data quality. The combined data set has 6,497 observations and 13 variables (11 physicochemical, *quality*, and *color*): 1,599 red wines (24.6%) and 4,898 white wines (75.4%). There are no missing values. However, 1,177 rows (18.1%; 240 red, 937 white) are exact duplicates of an earlier row. Because the file has no sample identifier, identical laboratory values can legitimately arise from different bottles or repeated analyses, so all rows were retained for the primary analyses (as the assessment requires the combined 6,497 observations); every key result was re-checked on the 5,320 unique rows as a sensitivity analysis.

Table 2

Descriptive statistics for the 11 physicochemical variables (combined data, N = 6,497)

Variable	*M*	*SD*	Min	Max	Skewness	Excess kurtosis	CV (%)
fixed acidity	7.22	1.30	3.80	15.90	1.72	5.06	18.0
volatile acidity	0.340	0.165	0.080	1.580	1.50	2.83	48.5
citric acid	0.319	0.145	0.000	1.660	0.47	2.40	45.6
residual sugar	5.44	4.76	0.60	65.80	1.44	4.36	87.4
chlorides	0.0560	0.0350	0.0090	0.6110	5.40	50.90	62.5
free sulfur dioxide	30.53	17.75	1.00	289.00	1.22	7.91	58.1
total sulfur dioxide	115.74	56.52	6.00	440.00	−0.00	−0.37	48.8
density	0.9947	0.0030	0.9871	1.0390	0.50	6.61	0.3
pH	3.22	0.16	2.72	4.01	0.39	0.37	5.0
sulphates	0.5313	0.1488	0.2200	2.0000	1.80	8.65	28.0
alcohol	10.49	1.19	8.00	14.90	0.57	−0.53	11.4

Note. CV = coefficient of variation (SD/mean × 100), used to compare variability across variables measured in different units.

Which variables are most variable? Raw standard deviations are not comparable because units differ (total sulfur dioxide has SD = 56.52 mg/dm³ whereas density has SD = 0.003 g/cm³), so relative variability is judged with the coefficient of variation. Residual sugar (CV = 87.4%), chlorides (62.5%), free sulfur dioxide (58.1%), total sulfur dioxide (48.8%), volatile acidity (48.5%), and citric acid (45.6%) are the most variable; density (0.3%), pH (5.0%), and alcohol (11.4%) are the most stable. Several variables are strongly right-skewed and heavy-tailed, most obviously chlorides (skewness = 5.40, excess kurtosis = 50.90), sulphates (1.80; 8.65), fixed acidity (1.72; 5.06), and residual sugar (1.44; 4.36).

Table 3

Means and standard deviations by wine colour, with standardised mean differences

Variable	Red *M* (*SD*)	White *M* (*SD*)	Cohen's *d* (White − Red)
fixed acidity	8.32 (1.74)	6.85 (0.84)	−1.29
volatile acidity	0.528 (0.179)	0.278 (0.101)	−2.00
citric acid	0.271 (0.195)	0.334 (0.121)	0.44
residual sugar	2.54 (1.41)	6.39 (5.07)	0.86
chlorides	0.0875 (0.0471)	0.0458 (0.0218)	−1.39
free sulfur dioxide	15.87 (10.46)	35.31 (17.01)	1.24
total sulfur dioxide	46.47 (32.90)	138.36 (42.50)	2.28
density	0.9967 (0.0019)	0.9940 (0.0030)	−0.98
pH	3.31 (0.15)	3.19 (0.15)	−0.81
sulphates	0.6581 (0.1695)	0.4898 (0.1141)	−1.29
alcohol	10.42 (1.07)	10.51 (1.23)	0.08

Note. Cohen's *d* uses the pooled SD; $|d| \approx 0.2$, 0.5, and 0.8 are conventionally small, medium, and large.

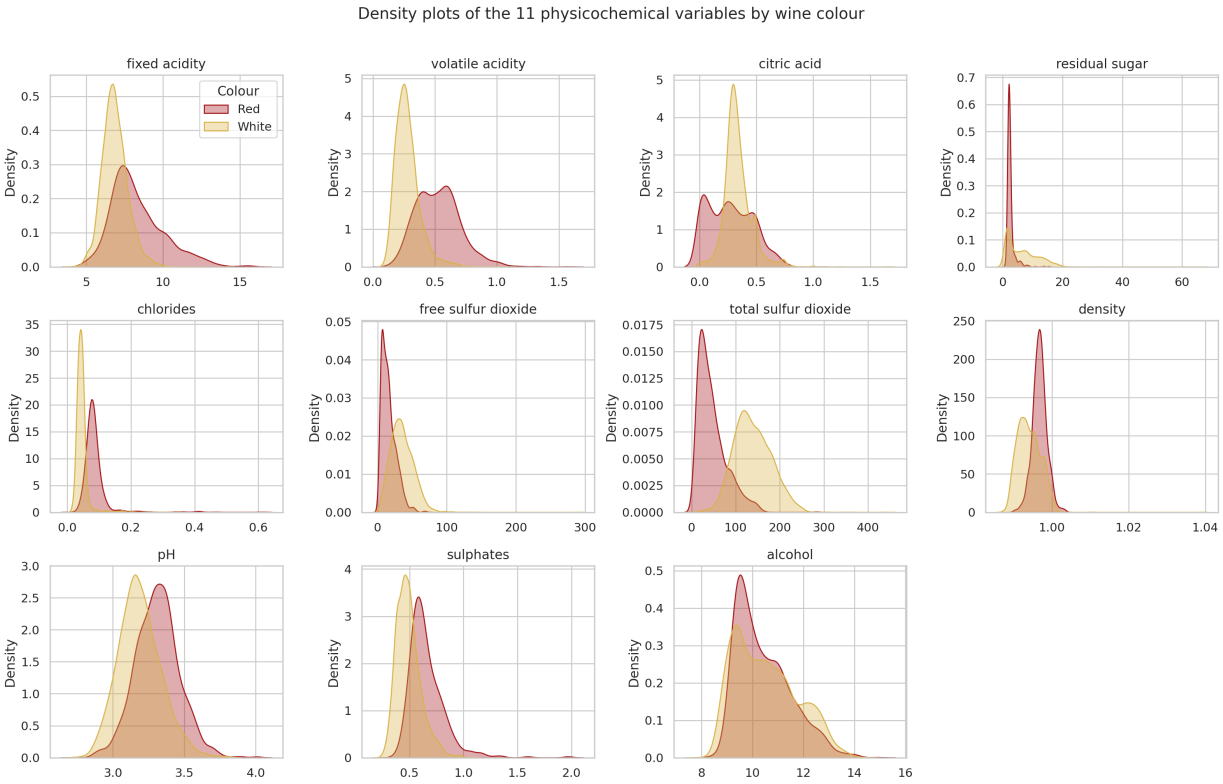


Figure 1. Density plots of the 11 physicochemical variables by wine colour (each density is scaled within colour).

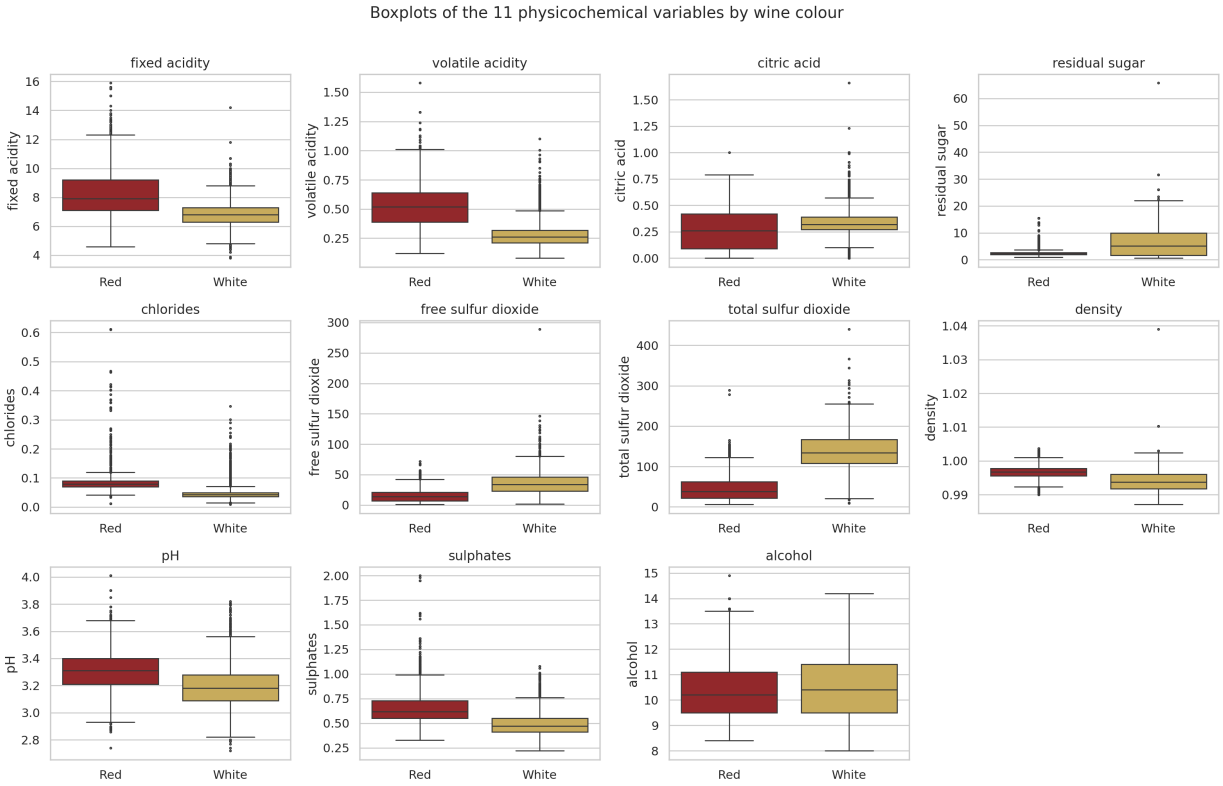


Figure 2. Boxplots of the 11 physicochemical variables by wine colour.

Differences between red and white wines. Differences are large for most variables (Table 3; Figures 1–2). White wines contain far more total sulfur dioxide (M = 138.36 vs. 46.47 mg/dm³, d = 2.28) and free sulfur dioxide (35.31 vs. 15.87, d = 1.24) and more residual sugar (6.39 vs. 2.54 g/dm³, d = 0.86). Red wines show higher volatile acidity (0.528 vs. 0.278, d = −2.00), chlorides (0.0875 vs. 0.0458, d = −1.39), fixed acidity (8.32 vs. 6.85, d = −1.29), sulphates (0.658 vs. 0.490, d = −1.30), density, and pH. Citric acid differs modestly (d = 0.44) and alcohol is practically identical (10.51% vs. 10.42%, d = 0.08).

Table 4

Pearson correlation matrix of the 11 physicochemical variables (N = 6,497)

Variable	1	2	3	4	5	6	7	8	9	10	11
1. fixed acidity	—										
2. volatile acidity	.22	—									
3. citric acid	.32	−.38	—								
4. residual sugar	−.11	−.20	.14	—							
5. chlorides	.30	.38	.04	−.13	—						
6. free sulfur dioxide	−.28	−.35	.13	.40	−.20	—					
7. total sulfur dioxide	−.33	−.41	.20	.50	−.28	.72	—				
8. density	.46	.27	.10	.55	.36	.03	.03	—			
9. pH	−.25	.26	−.33	−.27	.04	−.15	−.24	.01	—		
10. sulphates	.30	.23	.06	−.19	.40	−.19	−.28	.26	.19	—	
11. alcohol	−.10	−.04	−.01	−.36	−.26	−.18	−.27	−.69	.12	−.00	—

Note. $|r| \geq .04$ is significant at $p < .001$ with this sample size; interpret magnitude, not significance.

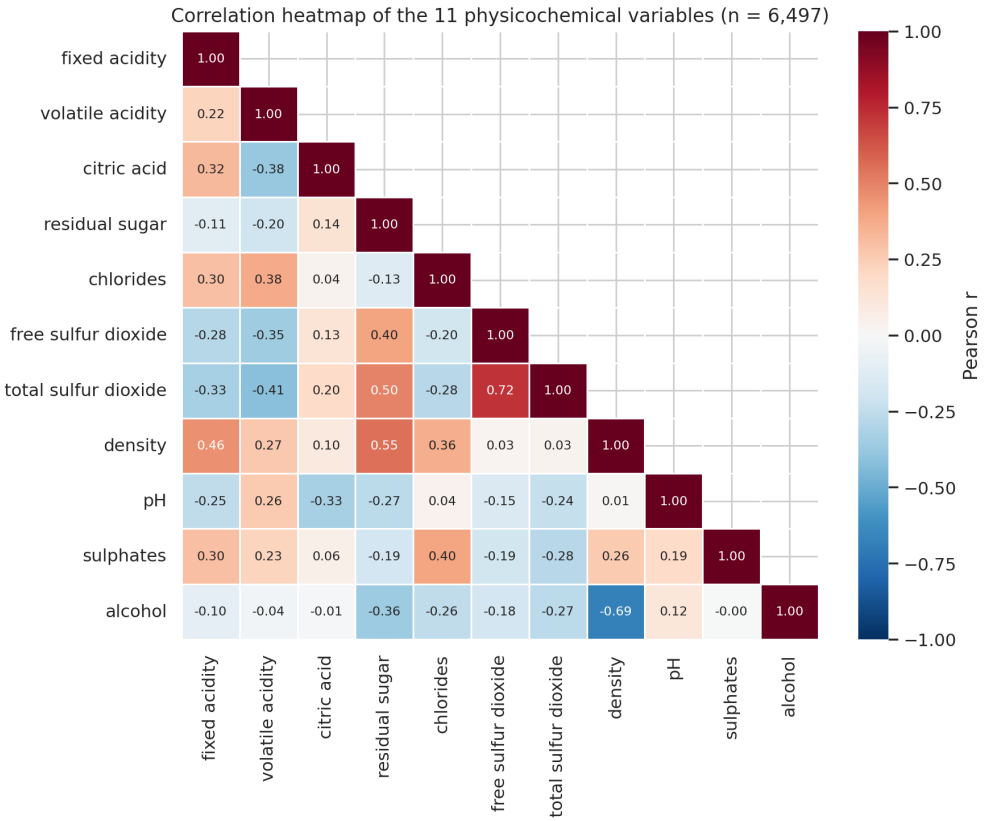


Figure 3. Correlation heatmap of the 11 physicochemical variables.

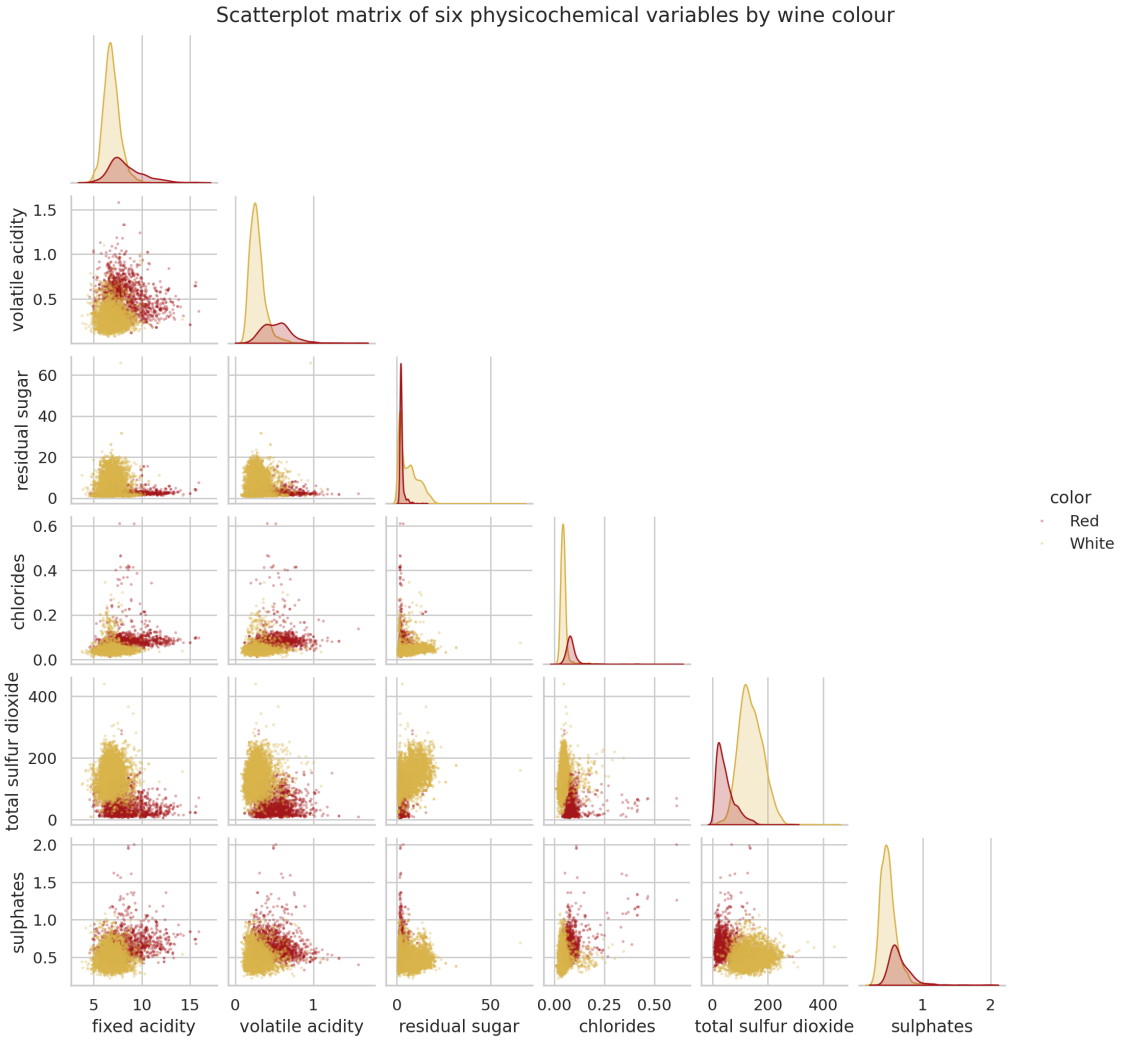


Figure 4. Scatterplot matrix of six physicochemical variables, coloured by wine colour (kernel densities on the diagonal).

Which variables are strongly correlated? Of the 55 variable pairs, 16 have $|r| \geq .30$ and two exceed .60: free and total sulfur dioxide ($r = .721$; free SO_2 is a component of total SO_2) and density and alcohol ($r = -.687$; ethanol is less dense than water). Next come residual sugar–density (.553), residual sugar–total SO_2 (.495), fixed acidity–density (.459), and volatile acidity–total SO_2 (−.414). Density is therefore tied to alcohol, sugar, and acids simultaneously, which anticipates the low KMO values reported below.

Outliers and unusual observations. Tukey's rule flags between 0.05% (density, alcohol) and 7.8% (citric acid) of observations per variable. The most extreme cases are in chlorides (133 values beyond 3 × IQR; maximum $|z| = 15.84$), density ($|z| = 14.77$, a single wine at 1.039 g/cm³), free sulfur dioxide (14.56), and residual sugar (a white wine with 65.8 g/dm³, $|z| = 12.69$). Multivariate screening (within-colour D^2 on all 11 variables; critical $\chi^2(11) = 31.26$) identified 197 wines (3.0%). These are plausible but unusual samples rather than obvious data-entry errors, so they were retained and their influence was tested in sensitivity analyses.

Considerations before MANOVA, PCA, and EFA. (1) Strong positive skew and heavy tails in chlorides, sulphates, residual sugar, free sulfur dioxide, and fixed acidity violate multivariate normality, which matters most for ML-based inference and is addressed with robust choices (Pillai's trace, permutation test, minres extraction). (2) Because the variables sit on very different scales, PCA and EFA must be based on correlations (standardised variables). (3) Density is almost a linear function of alcohol and sugar (VIF = 15.9), threatening singularity-like behaviour and Heywood cases in factor analysis. (4) Pooling the two colours creates mixtures (for example, bimodal total sulfur dioxide), so some correlations partly reflect between-colour differences. (5) The groups are imbalanced (3.1 : 1).

6.2 Multivariate Analysis of Variance (MANOVA)

Selection of dependent variables. I did not enter all variables automatically. Six variables were chosen on three criteria (Table 5): (a) *conceptual coverage* of distinct chemical domains: acidity (fixed and volatile acidity), sweetness (residual sugar), mineral/salt content (chlorides, sulphates), and preservation (total sulfur dioxide); (b) *non-trivial group separation* in the exploratory analysis ($|d| \geq 0.86$ for all six); and (c) *avoidance of redundancy*. Density was excluded because it is largely determined by alcohol and sugar (VIF = 15.9) and would make the within-group covariance matrix nearly singular; free sulfur dioxide was excluded because it is a component of total sulfur dioxide ($r = .72$) and separates the groups less well ($d = 1.24$ vs. 2.28); pH was excluded as a conceptual duplicate of the acidity measures; citric acid showed only a small difference ($d = 0.44$); and alcohol is practically identical across colours ($d = 0.08$). An eight-variable model that adds alcohol and density is reported as a sensitivity analysis.

Table 5

Evidence used to select the MANOVA dependent variables

Variable	Cohen's *d*	VIF (11 vars)	Selected	Rationale
fixed acidity	−1.29	4.90	Yes	Acidity domain
volatile acidity	−2.00	1.95	Yes	Acidity/spoilage domain
citric acid	0.44	1.61	No	Small effect
residual sugar	0.86	7.23	Yes	Sweetness domain
chlorides	−1.39	1.63	Yes	Salt domain
free sulfur dioxide	1.24	2.14	No	Redundant with total SO_2
total sulfur dioxide	2.28	2.94	Yes	Preservation domain; largest effect
density	−0.98	15.91	No	Redundant (alcohol, sugar)
pH	−0.81	2.54	No	Redundant with acids
sulphates	−1.29	1.54	Yes	Mineral/preservative domain
alcohol	0.08	4.78	No	Negligible effect

Assumption checks. Table 6 summarises the diagnostics; each is explained below.

Table 6

Summary of MANOVA assumption diagnostics

Assumption	Diagnostic	Result	Conclusion
Independence	Study design; duplicates	Observational; no producer/batch ID; 18.1% duplicate rows	Cannot be verified; limitation
Multivariate normality	Mardia skewness / kurtosis	Skewness $\chi^2(56)$ = 17,851 (red), 28,081 (white); kurtosis z = 195.9, 207.0; all $p <$ .001	Violated (heavy right tails); n is large; use Pillai + permutation
Multivariate outliers	Within-group D^2 ($\chi^2(6)$ at p = .001, cutoff 22.46)	176 wines (2.7%): 55 red, 121 white	Present; effect tested by removal
Equal covariance matrices	Box's M; $\lvert\Sigma\rvert$ ratio; Brown–Forsythe	M = 7,232.8, $\chi^2(21)$ = 7,222.4, $p <$ .001; $\lvert\Sigma_{white}\rvert/\lvert\Sigma_{red}\rvert$ = 0.184; all six BF tests $p <$ .001	Violated; Pillai's trace used; balanced re-test
Relationships among DVs	Within-group r; VIF	Max $\lvert r\rvert$ = .375 (within), .495 (pooled); VIF 1.22–1.68; $\det(\mathbf{R}_{within})$ = 0.747	Moderate; no multicollinearity

Independence. Independence cannot be demonstrated from the data. The wines are an observational sample of one region, and samples from the same producer, vintage, or batch may be more alike than random samples; the file has no identifier with which to model such clustering. The 1,177 duplicate rows hint at repeated entries or repeated analyses. Results are therefore interpreted as describing the samples at hand and are treated cautiously regarding inference to all Vinho Verde wines.

Multivariate normality. Mardia's tests reject normality in both groups for skewness and kurtosis (Mardia's kurtosis b_2 = 144.0 for red and 106.0 for white, against an expected 48 under normality). The chi-square Q–Q plots (Figure 5) bend sharply upward in the upper tail, driven by chlorides, residual sugar, and sulphates (group skewness up to 5.68). With 1,599 and 4,898 wines per group, the central limit theorem protects the sampling distribution of the mean vectors to a degree and multivariate tests of this kind are fairly insensitive to non-normality when group sizes are large, but heavy tails are the most damaging form of non-normality; I therefore added a permutation test and log-transformation check (below).

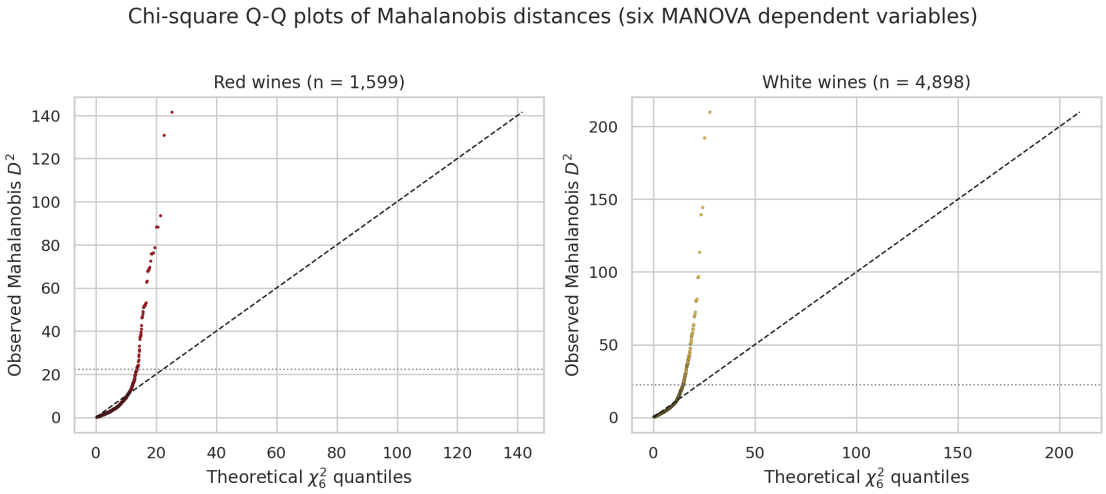


Figure 5. Chi-square Q–Q plots of within-group Mahalanobis distances for the six MANOVA dependent variables. Points above the dashed line indicate heavier tails than a multivariate normal distribution; the dotted line marks the p = .001 cutoff.

Outliers and homogeneity. 176 wines (2.7%) exceed the D^2 cutoff. Box's M is significant (p < .001) and the white wines' generalised variance is only 18.4% of the red wines'; variances

differ in opposite directions across variables (white wines vary much more in residual sugar and total SO_2, red wines vary more in fixed and volatile acidity, chlorides, and sulphates). Box's M is very sensitive with large samples, but the discrepancy is real and the group sizes are unequal, so Pillai's trace is used as the primary test and results are confirmed on a balanced random subsample (1,599 vs. 1,599), in which the test remains significant ($\chi^2(21) = 4{,}501.9$) yet equal group sizes make MANOVA far more robust to unequal covariance matrices.

Relationships among the dependent variables. Within-group correlations are modest (largest: residual sugar–total SO_2, $r = .375$), VIFs are 1.22–1.68, and the determinant of the within-group correlation matrix is 0.747. The variables are therefore related enough to justify a joint multivariate test but not so collinear that the test is unstable.

Table 7

Multivariate tests of the effect of wine colour on the six physicochemical variables

Test	Value	*F*	df_1	df_2	*p*	Partial η^2
Wilks' Λ	0.237	3,481.81	6	6,490	< .001	.763
Pillai's trace	0.763	3,481.81	6	6,490	< .001	.763
Hotelling–Lawley trace	3.219	3,481.81	6	6,490	< .001	.763
Roy's greatest root	3.219	3,481.81	6	6,490	< .001	.763

Note. With two groups there is a single discriminant dimension ($s = 1$), so all four statistics are transformations of one another and give the same exact *F*. Partial $\eta^2 = 1 - \Lambda$ = Pillai's *V*. Mahalanobis distance between group centroids $D = 4.165$; Hotelling's $T^2 = 20{,}906.98$.

MANOVA result. Red and white wines differ significantly on the combined set of six physicochemical variables, Pillai's $V = .763$, $F(6, 6490) = 3{,}481.81$, $p < .001$, partial $\eta^2 = .763$ (Wilks' $\Lambda = .237$). Colour accounts for about 76% of the multivariate variance in the six variables, an extremely large effect, and the group centroids are 4.16 pooled standard deviations apart in multivariate space. The permutation test agrees: the observed Pillai's *V* (.763) exceeds the largest of 10,000 permuted values (.005), giving $p < .001$ ($p \leq .0001$).

Table 8

Robustness of the MANOVA result to departures from the assumptions

Analysis	*N*	Pillai's *V*	*F*	*p*
Primary analysis	6,497	.763	3,481.8	< .001
log10-transformed DVs	6,497	.807	4,525.6	< .001
Multivariate outliers removed	6,321	.814	4,600.3	< .001
Duplicate rows removed	5,320	.756	2,744.3	< .001
Balanced subsample (1,599 vs 1,599)	3,198	.788	1,979.9	< .001
Eight-DV set (adds alcohol and density)	6,497	.853	4,692.2	< .001

Note. All analyses use the same six dependent variables except the last row (eight variables: adds alcohol and density). Group *F* has 6 and $N - 7$ degrees of freedom (8 and $N - 9$ in the last row).

Robustness. The multivariate effect is not an artefact of non-normality, outliers, duplicates, or imbalance: Pillai's *V* stays between .756 and .853 in every re-analysis (Table 8), and all *p*-values are below .001.

Table 9

Univariate follow-up ANOVAs for each dependent variable

Variable	Red *M* (*SD*)	White *M* (*SD*)	*F*(1, 6495)	η²	Cohen's *d*	Welch *F*
fixed acidity	8.32 (1.74)	6.85 (0.84)	2,016.5	.237	−1.29	1,051.2
volatile acidity	0.528 (0.179)	0.278 (0.101)	4,829.3	.426	−2.00	2,815.3
residual sugar	2.54 (1.41)	6.39 (5.07)	899.8	.122	0.86	2,285.1
chlorides	0.0875 (0.0471)	0.0458 (0.0218)	2,315.8	.263	−1.39	1,172.4
total sulfur dioxide	46.47 (32.90)	138.36 (42.50)	6,252.8	.490	2.28	8,077.0
sulphates	0.6581 (0.1695)	0.4898 (0.1141)	2,021.7	.237	−1.29	1,373.1

Note. All $p < .001$ (Bonferroni-adjusted $\alpha = .0083$). Welch's F does not assume equal variances and leads to the same conclusions. Post hoc pairwise tests are unnecessary because colour has only two levels: a significant effect is itself the Red vs. White contrast.

Table 10

Discriminant function separating white from red wines

Variable	Standardised coefficient	Structure *r*	Rank
fixed acidity	−1.53	−0.31	5
volatile acidity	−2.60	−0.48	2
residual sugar	0.08	0.21	6
chlorides	−1.17	−0.33	3
total sulfur dioxide	2.80	0.55	1
sulphates	−1.60	−0.31	4

Note. Positive values indicate higher scores for white wines. Structure r = correlation of the variable with the discriminant score (within groups).

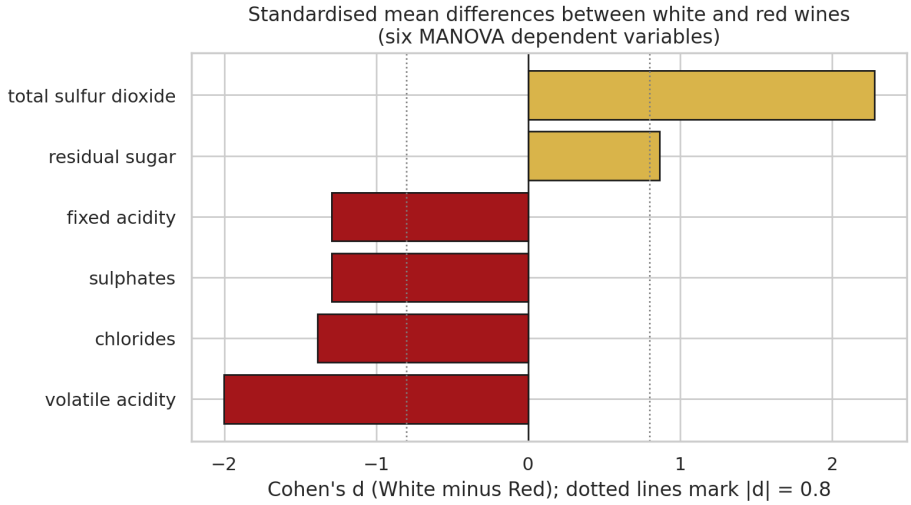


Figure 6. Standardised mean differences (Cohen's d, White − Red) for the six MANOVA dependent variables.

Follow-up analyses. All six univariate ANOVAs are significant after Bonferroni adjustment (Table 9). Total sulfur dioxide shows the largest effect ($\eta^2 = .490$, $d = 2.28$), followed by volatile acidity ($\eta^2 = .426$, $d = -2.00$), chlorides ($\eta^2 = .263$), sulphates and fixed acidity ($\eta^2 \approx .237$), and residual sugar ($\eta^2 = .122$). The discriminant function (Table 10) shows how the variables work *together*: total sulfur dioxide (structure $r = 0.55$) and volatile acidity ($r = -0.48$) define most of the separation, followed by chlorides, sulphates, and fixed acidity ($|r| \approx .31$–$.33$). Residual sugar differs strongly in the univariate test but adds almost nothing

uniquely (standardised coefficient = 0.08) because it overlaps with total sulfur dioxide within groups (r = .375). A linear discriminant classifier built from the six variables assigns wines to the correct colour with 97.8% 10-fold cross-validated accuracy, against 75.4% for always guessing “white.”

Interpretation in the context of wine chemistry. Red and white Vinho Verde wines are chemically distinct when the six characteristics are taken together: knowing a wine’s chemistry almost determines its colour. The white-wine profile is one of high sulfur dioxide, both free and total, and more residual sugar. The red-wine profile combines higher volatile and fixed acidity with more chloride and sulphate. The sulfur dioxide contrast is consistent with the widely documented practice of using more SO_2 to protect white wines from oxidation and microbial spoilage, since white wines lack the phenolic antioxidants that red wines extract from grape skins; the sugar contrast is consistent with white wines being more often left slightly sweet. These explanations are interpretive, because the data contain no information on winemaking practice. Alcohol does not discriminate, so ethanol content is not what separates the two types. Because nearly any difference is “significant” with 6,497 wines, the substantive conclusion rests on the effect sizes (partial η^2 = .763; $|d|$ between 0.86 and 2.28) and on the 97.8% classification accuracy rather than on the p-values.

6.3 Principal Component Analysis (PCA)

Correlation structure. The determinant of the correlation matrix is 0.0033 (far from 1), and Bartlett’s test rejects the identity-matrix hypothesis, $\chi^2(55) = 37{,}094.6$, $p < .001$. Sixteen of 55 correlations reach $|r| \geq .30$. The overall KMO measure, however, is only **.405**, below the conventional minimum of .50. The low value is concentrated in pH (MSA = .209), alcohol (.273), fixed acidity (.277), residual sugar (.294), and density (.305), variables whose relations are shared among several others (density, for instance, is jointly predicted by alcohol, sugar, and acidity), so partial correlations remain large relative to zero-order ones. PCA is a descriptive, variance-based technique that does not require a latent-variable model, so the data justify PCA, with the caution that the components will be broad composites rather than clean one-variable-one-component groupings.

Standardisation. The variables are measured in incommensurable units and on vastly different scales: total sulfur dioxide has a variance of about 3,195 mg^2/dm^6, whereas density’s variance is about 0.00001 g^2/cm^6. A covariance-matrix PCA would produce components determined almost entirely by total and free sulfur dioxide. Each variable was therefore converted to z-scores (mean 0, SD 1), so that PCA operates on the correlation matrix and every variable carries equal weight, with total variance = 11.

Table 11

PCA eigenvalues, variance explained, and retention criteria (first eight components)

Component	Eigenvalue	% variance	Cumulative %	PA 95th percentile	Broken-stick %	Retain?
PC1	3.030	27.5	27.5	1.081	27.5	Yes
PC2	2.494	22.7	50.2	1.060	18.4	Yes
PC3	1.556	14.1	64.4	1.044	13.8	Yes
PC4	0.971	8.8	73.2	1.031	10.8	No
PC5	0.720	6.5	79.7	1.019	8.5	No
PC6	0.607	5.5	85.2	1.008	6.7	No
PC7	0.523	4.8	90.0	0.997	5.2	No
PC8	0.501	4.6	94.6	0.986	3.9	No

Note. PA = parallel analysis (1,000 random-normal data sets with n = 6,497, p = 11). Components 9–11 have eigenvalues of 0.337, 0.228, and 0.033.

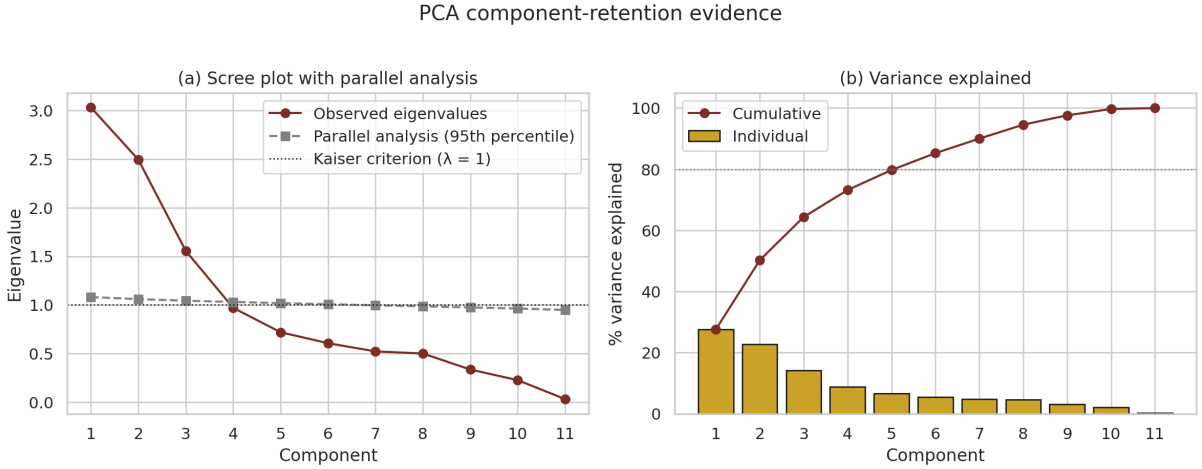


Figure 7. (a) Scree plot of the observed eigenvalues with the parallel-analysis threshold and Kaiser line; (b) individual and cumulative percentage of variance explained.

Number of components. Three lines of evidence agree on three components. (1) The first three eigenvalues (3.030, 2.494, 1.556) exceed both the Kaiser value of 1 and the parallel-analysis thresholds (≈ 1.04–1.08); the fourth (0.971) falls below both. (2) The broken-stick criterion also retains three. (3) The scree plot falls steeply through the third component, bends at the fourth, and flattens thereafter (successive differences of 0.54, 0.94, 0.59, then ≤ 0.25). Because eigenvalue>1 alone can be misleading, I weighed the borderline fourth component explicitly: its eigenvalue of 0.971 is practically 1, but it adds just 8.8 percentage points and defines a sulphates/pH pattern that is already weakly represented, so it was not retained in the interest of parsimony. Parallel analysis with this large n is a liberal criterion (any eigenvalue just above 1 is “significant”), which makes its agreement with the more conservative criteria reassuring.

Table 12

PCA component loadings (correlation-matrix PCA, three components retained)

Variable	PC1	PC2	PC3	Communality
fixed acidity	**−.416**	**.531**	**.542**	.749
volatile acidity	**−.663**	.186	−.383	.621
citric acid	.265	.289	**.737**	.697
residual sugar	**.602**	**.521**	−.205	.676
chlorides	**−.505**	**.498**	−.021	.503
free sulfur dioxide	**.750**	.114	−.167	.604
total sulfur dioxide	**.848**	.138	−.134	.757
density	−.078	**.922**	−.219	.905
pH	−.381	−.246	**−.568**	.528
sulphates	**−.512**	.303	.087	.361
alcohol	−.185	**−.734**	.326	.680
Eigenvalue	3.030	2.494	1.556	
% variance	27.5	22.7	14.1	64.4 (cum.)

Note. Loadings are correlations between variables and components; |loading| ≥ .40 in bold. Communality = proportion of each variable's variance reproduced by the three components. Signs are arbitrary; each component is oriented so its largest loading is positive.

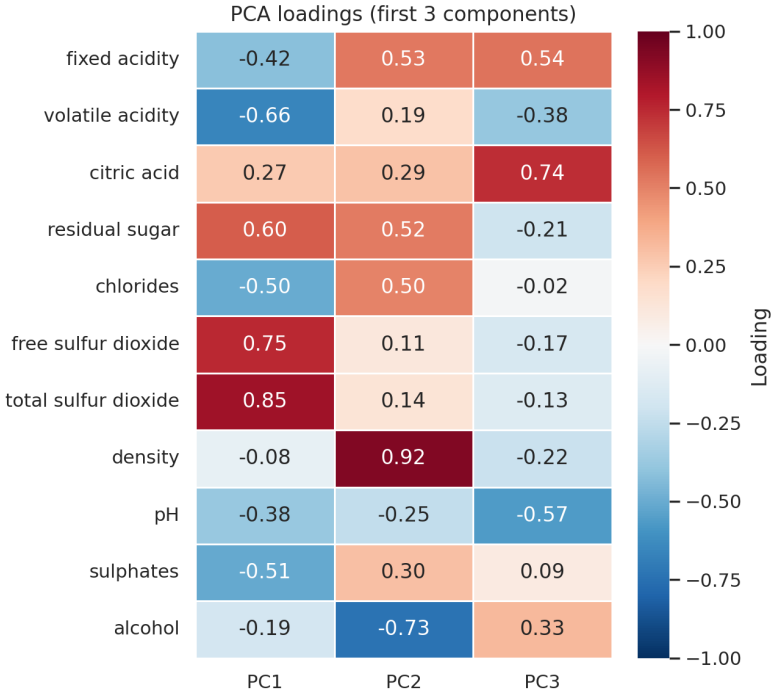


Figure 8. Heatmap of PCA loadings for the three retained components.

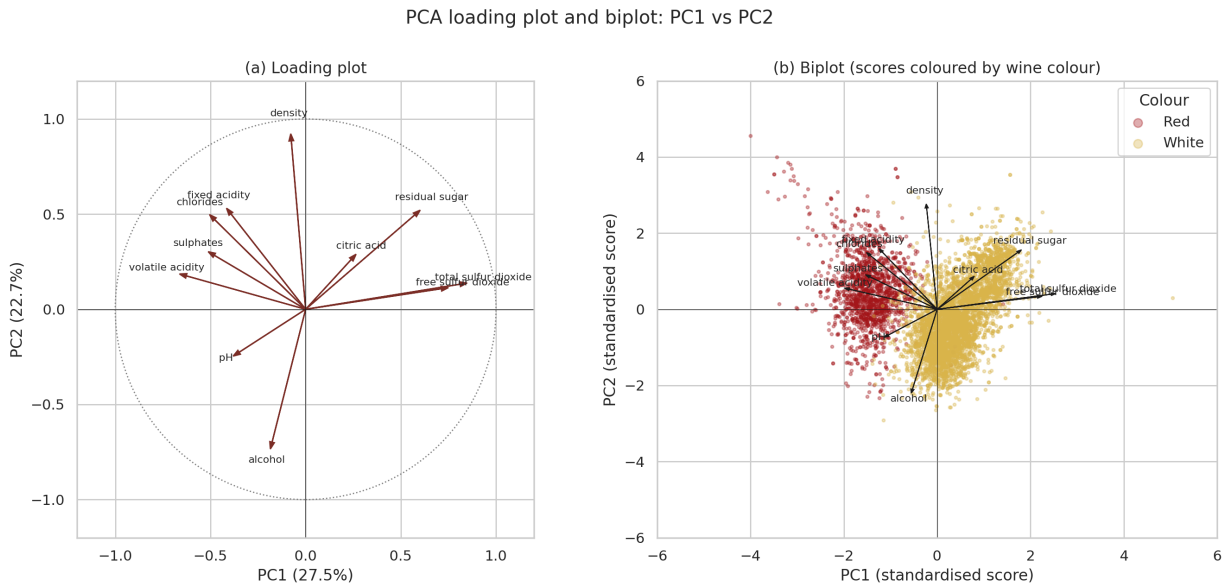


Figure 9. Loading plot (a) and biplot (b) for PC1 and PC2. In the biplot, points are wines coloured by colour and arrows are variable loadings.

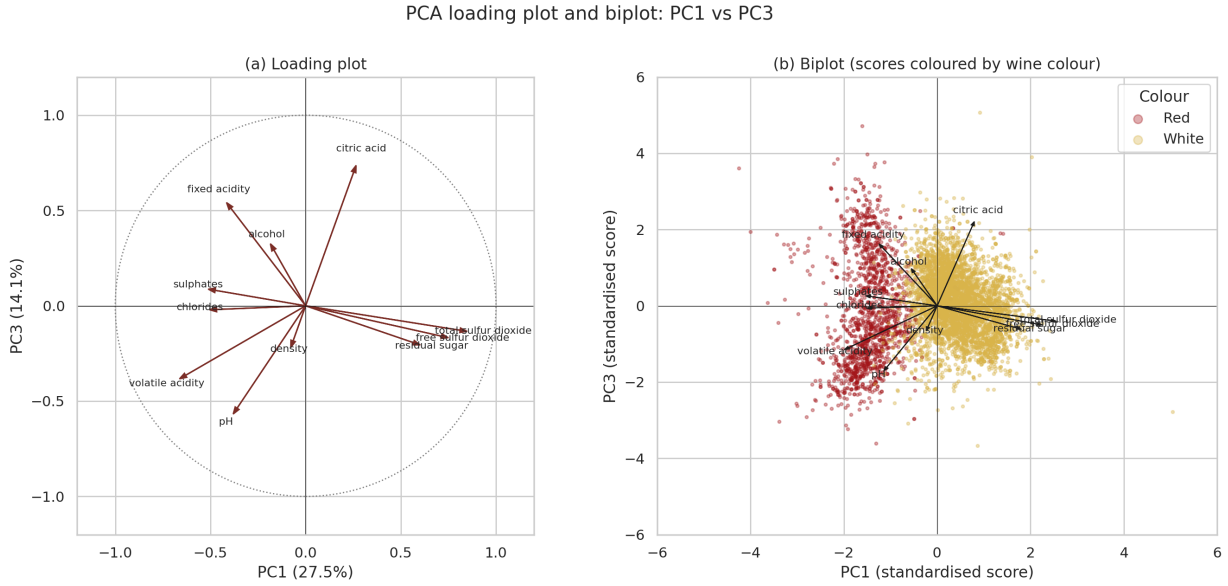


Figure 10. Loading plot (a) and biplot (b) for PC1 and PC3.

Interpretation of the components.

- **PC1 – Sulfur dioxide/preservation versus volatile acidity and salts (27.5% of variance).** Total SO_2 (.848), free SO_2 (.750), and residual sugar (.602) load positively; volatile acidity (−.663), sulphates (−.512), chlorides (−.505), and fixed acidity (−.416) load negatively. Wines high on PC1 are heavily sulfited and sweeter and low in volatile acidity and chlorides. White wines average +0.82 on PC1 and red wines −2.51 (Cohen's d = 3.39), so this component is essentially the red–white axis and mirrors the MANOVA discriminant function (total SO_2 and volatile acidity were the two leading discriminators).
- **PC2 – Density/alcohol (body and sweetness) dimension (22.7%).** Density (.922) and, with opposite sign, alcohol (−.734) dominate, with moderate contributions from fixed

acidity (.531), residual sugar (.521), and chlorides (.498). High scores describe dense, sugary, mineral wines with less alcohol; low scores describe light, alcoholic wines. This reflects the physical fact that dissolved solids raise density while ethanol lowers it. Colour matters little here (d = -0.68), so PC2 is a within-type style dimension.

- **PC3 – Acidity/freshness dimension (14.1%).** Citric acid (.737), fixed acidity (.542), and low pH (−.568), together with low volatile acidity (−.383), describe sharply acidic, “fresh” wines versus flatter, higher-pH wines. Colour is almost irrelevant here (d = 0.20).

Information retained. Three components are sufficient to summarise the 11 variables in the sense that they retain **64.4%** of the total standardised variance (27.5% + 22.7% + 14.1%) while reducing 11 variables to 3, but they are not a lossless summary: 35.6% of variance is discarded. Density (communality .905), total SO_2 (.757), fixed acidity (.748), citric acid (.697), alcohol (.680), and residual sugar (.676) are well represented, whereas sulphates (.361) and chlorides (.503) are represented poorly and would need a fourth component. The structure is stable: with duplicates removed PC1–PC3 explain 27.2%, 22.5%, and 14.4%; within red wines alone PC1 explains 28.2% and within white wines 29.3%, although PC2 shrinks to 17.5% and 14.3%, which shows that part of PC2’s variance in the pooled data comes from the colour mixture.

6.4 Exploratory Factor Analysis (EFA)

Factorability. The correlation matrix is shown in Table 4. Bartlett’s test of sphericity is highly significant, $\chi^2(55) = 37{,}094.6$, $p < .001$, so the variables share common variance. The overall KMO is .405, which is *unacceptable* by Kaiser’s (1974) benchmarks (≥ .50 required, ≥ .60 preferred); per-variable measures of sampling adequacy range from .209 (pH) to .751 (free SO_2) (Table 15). Five variables (pH, alcohol, fixed acidity, residual sugar, density) have MSA < .31. Dropping pH raises the KMO to .503, dropping pH and alcohol to .591, and dropping pH, alcohol, and density to .689. The data are therefore **only marginally suitable**: the correlations are strong enough overall for Bartlett’s test, but they are structured so that several variables are better predicted by combinations of other variables than by individual ones. As the assessment prescribes all 11 variables, I proceeded with the full set and treat the results as exploratory, flagging the weakly sampled variables when interpreting factors.

Number of factors. Tables 13–14 and Figure 11 bring together the evidence.

Table 13

Eigenvalues and parallel analysis for exploratory factor analysis

Factor	Common-factor eigenvalue	PA 95th percentile	PCA eigenvalue
1	2.632	0.084	3.030
2	2.276	0.063	2.494
3	1.153	0.046	1.556
4	0.469	0.033	0.971
5	0.401	0.021	0.720
6	0.204	0.009	0.607
7	0.161	−0.003	0.523
8	−0.066	−0.012	0.501

Note. Common-factor eigenvalues come from the correlation matrix with squared multiple correlations on the diagonal; factors 9–11 are negative.

Table 14

Fit and interpretability of one- to five-factor solutions

Factors	RMSR	% |residual| > .05	ML χ^2 (df)	RMSEA	TLI	MAP	Min. salient items per factor
1	.201	74.5	26,175 (44)	.302	.118	.0734	4
2	.102	58.2	13,964 (34)	.251	.392	.0723	5
3	.055	40.0	9,290 (25)	.239	.450	.0763	3
4	.040	20.0	5,452 (17)	.222	.525	.0976	2
5	.027	9.1	2,240 (10)	.185	.669	.1148	1

Note. RMSR from the minres solution; RMSEA and TLI from maximum-likelihood fits (supporting evidence only, because the data are non-normal); MAP = Velicer's average squared partial correlation after removing k components (minimum at k = 2 [.0723]; k = 3 gives .0763); salient = |promax pattern loading| ≥ .40.

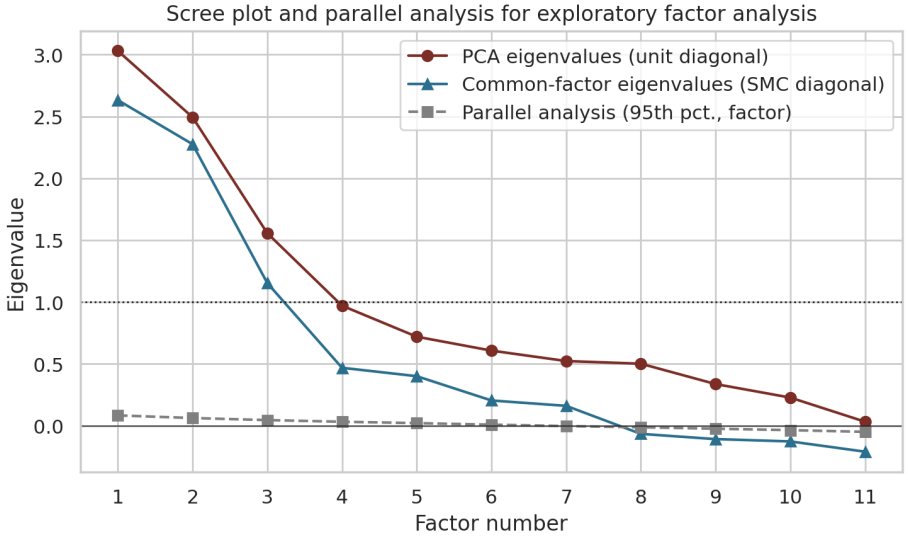


Figure 11. Scree plot comparing PCA eigenvalues, common-factor eigenvalues, and the parallel-analysis threshold.

The evidence is not unanimous, so the decision is explicit. Parallel analysis on the reduced correlation matrix suggests as many as 7 factors, but with n = 6,497 the random-data eigenvalues are essentially zero, so any non-trivial common variance is declared "significant"; the solutions with four or more factors produce factors defined by one or two variables (Table 14, last column) and unstable oblimin rotations, i.e., over-extraction. Velicer's MAP test is minimal at two factors, but the average partial correlation barely differs between two and three factors (.0723 vs. .0763), and two factors leave a large residual (RMSR = .102). The common-factor scree shows a clear elbow after the third factor (1.153 → 0.469), the Kaiser criterion and PCA parallel analysis both give three, the three-factor solution reduces RMSR to .055, and only the three-factor solution has at least three salient indicators for every factor. I therefore retained **three factors**. Absolute model fit is not good at any size (RMSEA ≥ .185 even with five factors, TLI ≤ .67), which is consistent with the non-normal distributions, the low KMO, and the near-deterministic role of density, and the three-factor solution should be viewed as a useful approximation rather than a well-fitting measurement model.

Extraction and rotation. Factors were extracted with the minimum-residual (minres) method. Unlike maximum likelihood, it does not rely on multivariate normality, which

Mardia's tests and the skewness values show to be badly violated, and unlike PCA it models common variance only, which is what the research question asks (Fabrigar et al., 1999). Rotation was **direct oblimin (oblique)**. Chemical dimensions such as preservation, density, and acidity have no reason to be uncorrelated, and an oblique rotation lets the data decide: the estimated factor correlations are $\varphi_{12} = -.020$, $\varphi_{13} = .006$, and $\varphi_{23} = .171$ (maximum $|\varphi| = .171$). Because these are small (below .32), the factors are nearly orthogonal, and the oblique solution loses little parsimony. A varimax solution assigns the same variables to the same factors but differs in magnitudes (Tucker congruence with oblimin: 0.91, 0.92, 0.93), so oblimin is retained as the primary solution.

Table 15

EFA pattern loadings, communalities, and uniquenesses (minres extraction, oblimin rotation, three factors)

Variable	F1	F2	F3	Communality (h^2)	Uniqueness (u^2)	MSA
fixed acidity	**−.434**	.274	**.613**	.699	.301	.277
volatile acidity	**−.549**	.338	−.319	.490	.510	.613
citric acid	.188	−.064	**.692**	.505	.494	.622
residual sugar	**.597**	**.451**	.019	.552	.448	.294
chlorides	−.394	.392	.052	.325	.675	.729
free sulfur dioxide	**.699**	.071	−.024	.491	.509	.751
total sulfur dioxide	**.860**	.084	.007	.745	.255	.718
density	.002	**1.012**	.044	1.041	−.041	.305
pH	−.278	.030	**−.466**	.292	.708	.209
sulphates	−.398	.216	.082	.221	.779	.564
alcohol	−.232	**−.662**	.052	.477	.523	.273
SS loadings (variance)	2.556	2.083	1.199			
% of total variance	23.2	18.9	10.9	53.1 (cum.)		

Note. |Pattern loading| ≥ .40 in bold. MSA = per-variable KMO. Factor correlations: F1–F2 = −.020, F1–F3 = .006, F2–F3 = .171. Residual fit: RMSR = .055; 40% of off-diagonal residuals exceed .05. Density's communality (1.041) is a Heywood case (see text).

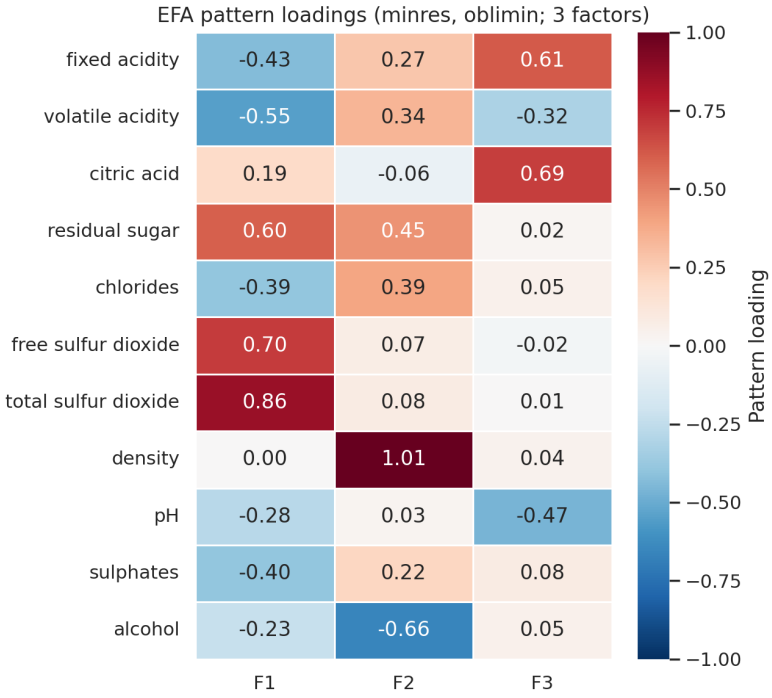


Figure 12. Heatmap of the oblimin-rotated EFA pattern loadings (three factors).

Loadings, communalities, and uniqueness.

- **Strong loadings:** density on F2 (1.012), total SO_2 on F1 (.860), free SO_2 on F1 (.699), citric acid on F3 (.692), alcohol on F2 (−.662; opposite sign to density), residual sugar on F1 (.597) and F2 (.451), and fixed acidity on F3 (.613).
- **Weak variables:** sulphates (highest loading .398; h^2 = .221), pH (.466; h^2 = .292), and chlorides (.394 on F1 and .392 on F2; h^2 = .325) are explained poorly by the three factors, with 68–78% of their variance unique, which agrees with the PCA communalities and the low MSAs.
- **Cross-loadings:** volatile acidity loads on all three factors (F1 −.549, F2 .338, F3 −.319); fixed acidity on F1 (−.434) and F3 (.613); residual sugar on F1 (.597) and F2 (.451); and chlorides almost equally on F1 and F2. The factors are therefore not “simple structure”: several variables are chemically linked to more than one dimension.
- **Heywood case:** density’s estimated communality is 1.041 (uniqueness −.041), an improper solution that signals density is reproduced almost perfectly from the other variables; this is expected, because density is physically determined by alcohol, sugar, and dissolved solids. The overshoot is small and affects no other variable, so density was retained as a marker of F2 but its loading should not be taken at face value.

Interpretation of the factors. The three factors jointly explain 53.1% of the total variance (23.2%, 18.9%, and 10.9%).

- **Factor 1 – Sulfur dioxide/preservation (“wine style”) factor.** Total SO_2 (.860), free SO_2 (.699), and residual sugar (.597) load positively, whereas volatile acidity (−.549), fixed acidity (−.434), sulphates (−.398), and chlorides (−.394) load negatively. It contrasts sulfite-protected, sweeter wines with wines high in volatile acidity and mineral content, which is the white–red contrast also found by MANOVA and PC1.

- **Factor 2 – Density/alcohol (body and sweetness) factor.** Density (1.012) and alcohol (−.662) are its core, with residual sugar (.451) and chlorides (.392) as secondary contributors: wines in which dissolved sugar and solids raise density and ethanol lowers it.
- **Factor 3 – Acidity factor.** Citric acid (.692), fixed acidity (.613), and pH (−.466; acidic wines have low pH), with volatile acidity (−.319), describe the strength of non-volatile acidity and its pH expression.

Overall, the EFA reveals three latent dimensions of Vinho Verde chemistry: **preservation/sulfur dioxide, body (density–alcohol–sugar), and acidity**, each supported by at least three observed variables with salient loadings. Sulphates, chlorides, and pH are only weakly captured and more likely reflect additional influences not modelled here.

Sensitivity to pooling. When the colour means are subtracted (within-colour data), a three-factor solution reproduces F3 almost exactly (congruence = 0.99) and F2 well (0.93), but F1 only partially (0.71). The sulfur dioxide/preservation factor is thus partly a between-colour construct, whereas the body and acidity factors describe variation that exists within both wine types.

7. Integrated Discussion

MANOVA: what distinguishes red from white wines? The six-variable chemical profile separates the colours almost perfectly (Pillai's V = .763; 97.8% cross-validated classification). The decisive variables are total sulfur dioxide (higher in white) and volatile acidity (higher in red); chlorides, sulphates, and fixed acidity (all higher in red) add a smaller share, and residual sugar (higher in white) is largely redundant with sulfur dioxide once the other variables are controlled. Alcohol does not discriminate.

PCA: what are the major dimensions of variation? Three components retain 64.4% of the variance: PC1 (sulfur dioxide/preservation versus volatile acidity and salts), PC2 (density versus alcohol, with sugar), and PC3 (acidity/pH, citric acid). Only PC1 is strongly tied to colour (d = 3.39), whereas PC2 (d = −0.68) and PC3 (d = 0.20) mostly describe variation within both types.

EFA: what latent factors explain the relationships? Three correlated-but-nearly-orthogonal factors, preservation, body, and acidity, explain 53.1% of the total variance and the correlations among the variables, with density, total sulfur dioxide, and alcohol among the best-defined indicators and sulphates, chlorides, and pH among the worst.

Table 16
Comparison of the PCA and EFA solutions

Aspect	PCA	EFA
Purpose	Composite summary of total variance	Model of shared (common) variance via latent factors
Diagonal / error	Unit diagonal; no error term	Communalities on the diagonal; unique variance modelled
Variance explained (3 dimensions)	64.4% of total variance	53.1% of total variance (common variance only)
Dimension 1	PC1: SO_2, sugar vs. volatile acidity, sulphates, chlorides	F1: SO_2, sugar vs. volatile and fixed acidity, sulphates, chlorides
Dimension 2	PC2: density, alcohol (−), fixed acidity, sugar, chlorides	F2: density, alcohol (−), sugar, chlorides
Dimension 3	PC3: citric acid, fixed acidity, pH (−), volatile acidity (−)	F3: citric acid, fixed acidity, pH (−), volatile acidity (−)
Inter-dimension correlation	Zero by construction	Estimated (oblimin): $\|\varphi\| \leq .171$
Weakly represented variables	Sulphates (.361), chlorides (.503), pH (.528)	Sulphates (.221), pH (.292), chlorides (.325)
Problems encountered	None; KMO caveat	Heywood case (density); poor absolute fit; low KMO

Comparison of PCA and EFA. The two techniques tell the same three-dimensional story (Table 16). Because PCA analyses total variance and EFA only common variance, the PCA components explain more variance (64.4% vs. 53.1%) and load higher on variables with large unique variance, while the EFA reveals that this extra variance is partly unique to individual variables, mostly sulphates, pH, and chlorides. The EFA is also more demanding of the data, as shown by the Heywood case and weak fit, whereas the PCA is always well defined; this is in keeping with the low KMO, which warns that a common-factor model fits these data less naturally than a descriptive summary. The variables that matter most in *both* analyses are total sulfur dioxide, free sulfur dioxide, density, alcohol, residual sugar, citric acid, and fixed acidity, each with substantial loadings on the same dimension in both solutions.

Substantive interpretation. The chemistry of Vinho Verde wines is organised around three largely independent themes: how the wine is protected (sulfur dioxide) and whether it is sweet, how heavy and alcoholic it is (the density–alcohol–sugar balance), and how acidic it is. Wine colour is predominantly tied to the first theme (sulfur dioxide, volatile acidity, and salts), consistently across MANOVA, PCA (PC1, $d = 3.39$), and EFA (F1), whereas body and acidity vary in both red and white wines. A producer or analyst can therefore think in terms of about three summary measurements, with sulphates, chlorides, and pH retaining individual information that the three dimensions do not capture.

Limitations.

- **Observational, non-independent data.** The wines are not a random sample and no producer, vintage, or batch identifier is available; independence, required by MANOVA, cannot be established, and 18.1% of rows are exact duplicates. Conclusions describe these samples and cannot be generalised without caution.
- **Imbalanced groups (3.1 : 1) and unequal covariance matrices.** Box’s M indicates unequal covariance matrices. Pillai’s trace and a balanced-subsample check mitigate but do not remove the concern.

- **Non-normality and outliers.** Heavy right tails (chlorides, sulphates, residual sugar) and 2.7–3.0% multivariate outliers violate normality assumptions. The conclusions survive transformation and removal, but PCA/EFA loadings are sensitive to such observations, and I did not transform the variables for PCA/EFA.
- **Marginal factorability and weak model fit.** The overall KMO (.405) is unacceptably low, density produced a Heywood case, and the three-factor solution fits only moderately (RMSR = .055; ML RMSEA = .239). The factors are exploratory approximations and should be confirmed in an independent sample (for example, with a confirmatory factor analysis).
- **Pooling colours and interpretation of latent factors.** Combining red and white wines makes the factor structure partly reflect colour (F1 within-colour congruence = .71), and the labels “preservation,” “body,” and “acidity” are interpretations of loading patterns, not measured constructs. Sensory quality, which the assessment excludes, may also relate to these dimensions, but that relation was not analysed.
- **Statistical versus practical significance.** With N = 6,497, all tests are significant; effect sizes and cross-validated accuracy are the appropriate indicators of importance.

8. Conclusion

Red and white Vinho Verde wines differ markedly in their joint physicochemical profile (Pillai’s V = .763, $F(6, 6490)$ = 3,481.81, $p < .001$; partial η^2 = .763), with total sulfur dioxide and volatile acidity as the main discriminators and a 97.8% cross-validated classification accuracy; the result holds under transformation, outlier removal, duplicate removal, balancing, and permutation testing. The 11 measurements can be summarised by three principal components that retain 64.4% of the variance and are interpretable as sulfur dioxide/preservation, density–alcohol body, and acidity dimensions; an oblique three-factor model reaches the same structure with 53.1% common variance. The suitability of the data for factor analysis is marginal (KMO = .405) and sulphates, chlorides, and pH remain poorly explained, so the latent-factor results should be treated as exploratory. Overall, the wines’ chemistry is organised in a small number of interpretable dimensions, one of which closely tracks wine colour.

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